

Pictures Become Numbers

Project: Helpful Classroom Sorter

1. Your mission

Your learning target

I can encode and reconstruct a small image using an explicit grid convention.

I can explain why pixel counts alone cannot recover an arrangement.

I can distinguish numerical image data from a model's interpretation of that data.

Remember before reading

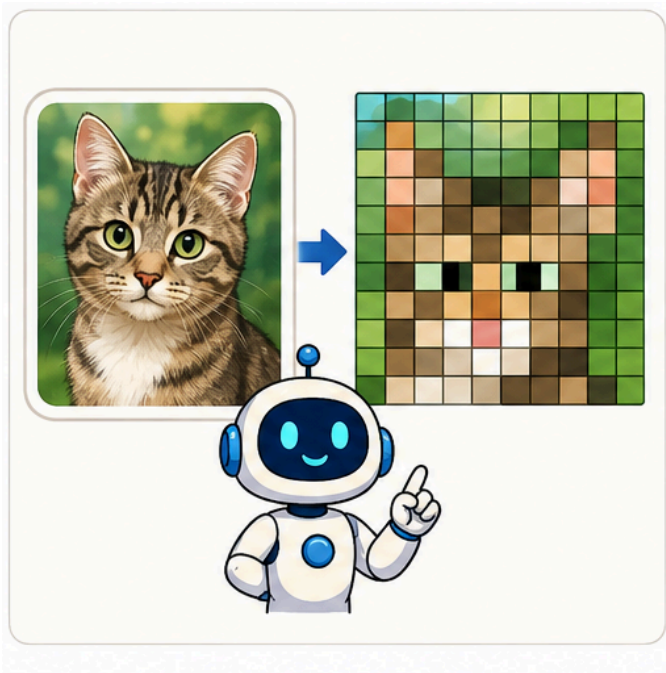
Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Input, Process, Output. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

A digital image is a grid of measured color values.



LEARNING TARGET Encode a 2×3 monochrome image with top row black–white–black and bottom row white–black–white, using 1 for black. A partner reads columns first. Explain why the image changes although the six values are identical.

Learner name:

Readiness for the connected explanation

Two thresholds can agree on all fitting examples but disagree on a new input. Does zero fitting error identify one uniquely correct rule?

This week's end goal

By the end, I can encode pixel positions and show why a count cannot reconstruct a picture.

2. Notice the evidence

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; evidence tells us what the clues mean.



MYSTERY A 5 by 5 black-and-white picture can be stored as 25 numbers, one value for each pixel position.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first prediction and state the clue that supports it.

My observations, prediction, and evidence:

GOOD THINKING A prediction can change when stronger evidence appears.

A closer explanation

Imagine sending a small black-and-white sign using only numbers. We can divide the sign into a grid, record 1 for a black square and 0 for a white square, and read each row from left to right, starting at the top. The receiver needs the width, height and these conventions to reconstruct the same arrangement.

A pixel is a position with a recorded value. In our simplified binary image, one value selects black or white. Other image formats can record brightness levels or multiple colour channels. The convention 1 means black belongs to this exercise; it is not a universal rule for every digital image.

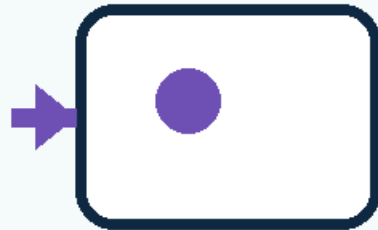
Encoding a picture does not recognise what it depicts. A program can store or display the numbers without calling the image a letter, an arrow or a person. Interpretation is another operation, just as displaying a book title was different from reading the entire book in Week 2.

3. See how it works

VISUAL MECHANISM

Input → **Process** → **Output**

Follow the information instead of imagining magic inside the machine.



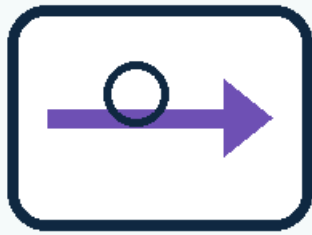
INPUT

light measured at grid locations



PROCESS

store and transform pixel values



OUTPUT

a displayed, resized, blurred, or classified image

INPUT light measured at grid locations

PROCESS store and transform pixel values

OUTPUT a displayed, resized, blurred, or classified image

CORE IDEA A digital image is a grid of measured color values.

UNDER THE HOOD Digital images sample a scene across a spatial grid. Each pixel stores channel values, commonly red, green, and blue. Resolution controls sampling density; transformations operate on numbers, not on an intact object inside the computer.

Draw a fresh example with all three stages:

A closer explanation

A three by three binary image

Read rows top to bottom. Within each row read left to right. 1 is black and 0 is white.

0 · 1 · 0

0: 1

1: 1

0: 1

0: 0

1: 1

0: 0

The image has width 3 and height 3, so it contains 9 pixels. Five are black. The row records preserve where those five black pixels occur.

The first row is 0,1,0, so only its middle pixel is black. The second row is 1,1,1, so all three are black. The third row repeats the first. This reconstructs the displayed cross-like pattern under our convention.

A flattened record would be 0,1,0,1,1,1,0,1,0. Without the grid dimensions, the same nine values could also be drawn in one row of nine. Keep the shape information with the values. Width times height counts pixels; it does not tell us their contents.

Connect the picture and the explanation

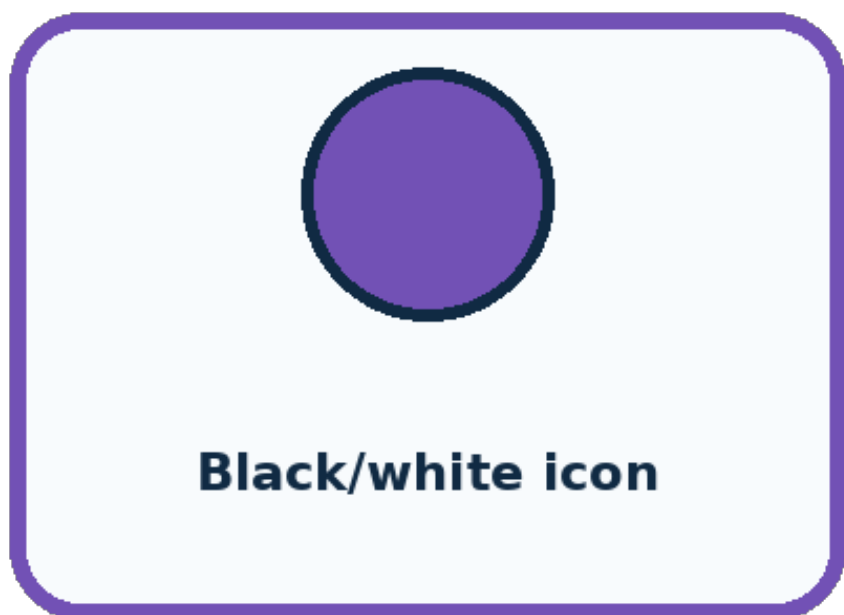
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

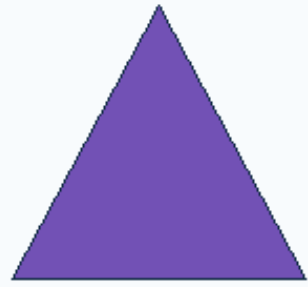
4. Compare the cases

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the evidence carefully.





Gray image



Color image



CASE 1 — Black/white icon Clue: 0 and 1 values. Meaning: two possible brightness levels.

CASE 2 — Gray image Clue: one brightness number per pixel. Meaning: many light levels.

CASE 3 — Color image Clue: red, green, blue values. Meaning: three channels.

CASE 4 — Blur Clue: nearby values are mixed. Meaning: detail is reduced.

BOUNDARY A pixel is a measurement at a location, not a tiny copy of the object.

Which two pictures are easiest to confuse? What evidence separates them?

Change one thing in the pictured case

Change: Change A's bottom-left pixel from 0 to 1.

Prediction: A now has four dark pixels; its row encoding is 101 / 110.

Reason: Exactly one light pixel becomes dark; the other five positions remain unchanged.

5. Words that unlock the idea

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.



pixel



grid



channel



resolution

Word · Meaning · My example

Word: pixel

Meaning: one picture element in a grid

My example: _____

Word: grid

Meaning: rows and columns that locate values

My example: _____

Word: channel

Meaning: one measured color component

My example: _____

Word: resolution

Meaning: number of pixels used to represent an image

My example: _____

Misconception detective

CLAIM TO REPAIR The computer stores a real cat inside the image file.

Use one vocabulary word and one piece of evidence to repair it:

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word · Meaning

Word: Encoding

Meaning: A convention for representing information with symbols or numbers

Word: Spatial arrangement

Meaning: Where elements are located relative to one another

Word: Interpretation

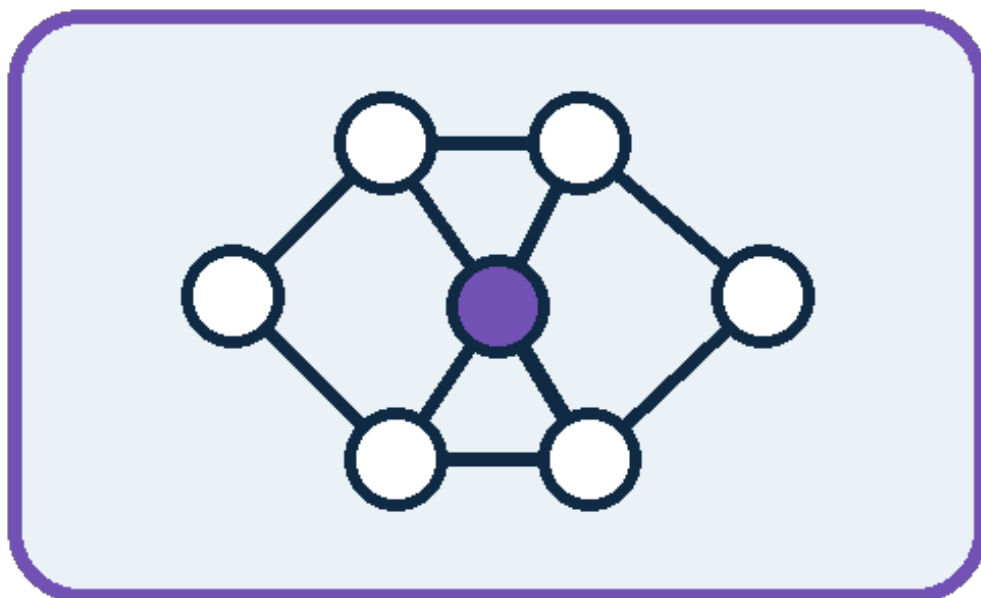
Meaning: A meaning or category assigned to a representation using a rule, model or human judgment

6. Follow a worked example

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable evidence.



CASE A 5 by 5 black-and-white picture can be stored as 25 numbers, one value for each pixel position.

STEP 1 — OBSERVE light measured at grid locations

STEP 2 — TRACE store and transform pixel values

STEP 3 — CONCLUDE a displayed, resized, blurred, or classified image

MODEL REASONING Each grid position becomes one displayed square. A blur combines nearby values, so isolated corners and thin lines disappear before large regions. The computer transforms numbers; the human viewer recognizes the shape.

DO NOT OVERCLAIM A pixel is a measurement at a location, not a tiny copy of the object.

What new evidence could change this answer?

A closer explanation

Suppose we discard the row records and retain only black pixel count=5. That count matches the cross, but it also matches rows 1,1,1 then 1,1,0 then 0,0,0. The second arrangement fills the top row and two positions below it. Both have five black pixels and nine pixels in total, yet they are different images.

Stored information · What it establishes · What remains unknown

Stored information: Width 3 and height 3

What it establishes: There are 9 positions

What remains unknown: The value at each position

Stored information: Black pixel count 5

What it establishes: Five positions are black under this code

What remains unknown: Which five positions

Stored information: Dimensions and all three ordered rows

What it establishes: The full binary grid under this convention

What remains unknown: What a viewer or model will call the picture

A pixel count can still be a useful feature for some tasks. For example, it can separate empty grids from grids with at least one mark. But it cannot separate every shape. To judge a feature, ask whether it retains the distinctions required by the task.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Pictures Become Numbers

Two fictional 2-by-3 pixel badges contain the same number of dark squares.

The facts we will use

Dark=1, light=0; rows are read top to bottom, left to right.

Badge A rows: 1,0,1 / 0,1,0. Badge B: 1,1,1 / 0,0,0.

Pixel positions: two badges, same dark count

	Left	Middle	Right
A top	1	0	1
A bottom	0	1	0
B top	1	1	1
B bottom	0	0	0

Read a cell using both its row and column labels.

Pictures Become Numbers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Pixel convention

Each image has two rows and three columns; 1 means dark.



2. Badge A

Top 101, bottom 010: three dark pixels.



3. Badge B

Top 111, bottom 000: also three dark pixels.



4. Compare arrangements

The totals agree, but middle-top and middle-bottom positions differ.

CHANGE ONE THING

Change A's bottom-left pixel from 0 to 1.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Pixel convention

What happens and why: Each image has two rows and three columns; 1 means dark.

Step: 2. Badge A

What happens and why: Top 101, bottom 010: three dark pixels.

Step: 3. Badge B

What happens and why: Top 111, bottom 000: also three dark pixels.

Step: 4. Compare arrangements

What happens and why: The totals agree, but middle-top and middle-bottom positions differ.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A grid preserves where each brightness belongs. Adding the six values preserves only the dark count, which is 3 in both images. The total therefore cannot distinguish these badges. Dimensions and reading order belong to the representation, not to an optional caption.

What this example does not establish

This toy binary encoding does not include real photographs' colour channels, compression or illumination effects.

Read the labelled picture: Find a position where the original badges differ, then read the changed A back as a picture.

Change one thing: Change A's bottom-left pixel from 0 to 1. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case · Evidence · What it shows

Case: Black/white icon

Evidence: 0 and 1 values

What it shows: two possible brightness levels

Case: Gray image

Evidence: one brightness number per pixel

What it shows: many light levels

Case: Color image

Evidence: red, green, blue values

What it shows: three channels

Case: Blur

Evidence: nearby values are mixed

What it shows: detail is reduced

Repair two tempting claims

CLAIM A The computer stores a real cat inside the image file.

Repair:

CLAIM B More pixels always guarantee a correct classification.

Repair:

STRONG ANSWER FRAME I claim _____ because the evidence shows _____. The process is _____. One limit is _____.

A closer explanation

Higher resolution offers more positions for recording detail, but it does not guarantee a useful image. A larger blurry image can still hide a feature, and a cropped image can omit it entirely. A model's label also depends on what it was fitted to distinguish and the conditions represented in its examples.

Keep claims at the right stage. These nine values encode this grid is a reconstruction claim. This grid depicts a plus sign is an interpretation. This sign tells a person what to do adds context and authority. One does not automatically establish the others.

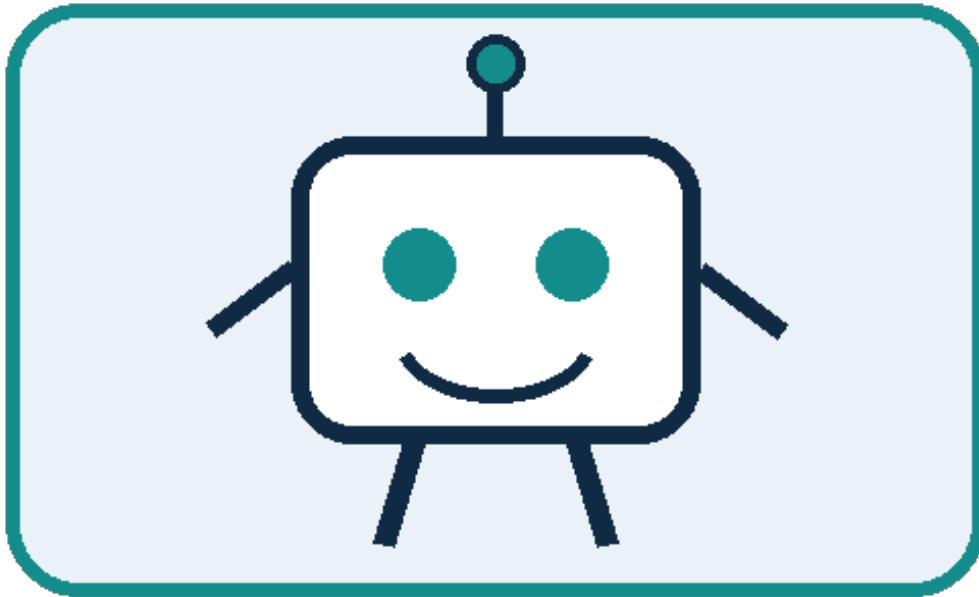
Can two different three by three images have the same black-pixel count? Explain with a specific arrangement change

8. Practise together

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A digital image is a grid of measured color values.

Use a fresh example to explain pixel.

Trace a new input → process → output example.

Find and repair the error in: More pixels always guarantee a correct classification.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the evidence. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Use width 3, height 2 and rows 1,0,1 then 0,1,0. Draw two rows of three squares and fill exactly the positions marked 1. Check by reading your drawing back into numbers. A round trip from code to drawing to code can reveal a swapped row or a missed pixel.

Draw the two by three image, write its black-pixel count and explain why saying only three black pixels would be insufficient

If an image classifier mislabels a picture, first inspect the input representation. Was the width correct? Were the rows reversed? Did the program use the intended colour convention? A later classifier cannot reliably repair every mistake in how its input was prepared.

Practise with less help: Preserve the missing field

Use workbook W2 for the connected example “Images as numerical grids”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Images as numerical grids. Proposed explanation: “The number one always means black”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Find a position where the original badges differ, then read the changed A back as a picture.

My explanation using a pictured step and the given facts

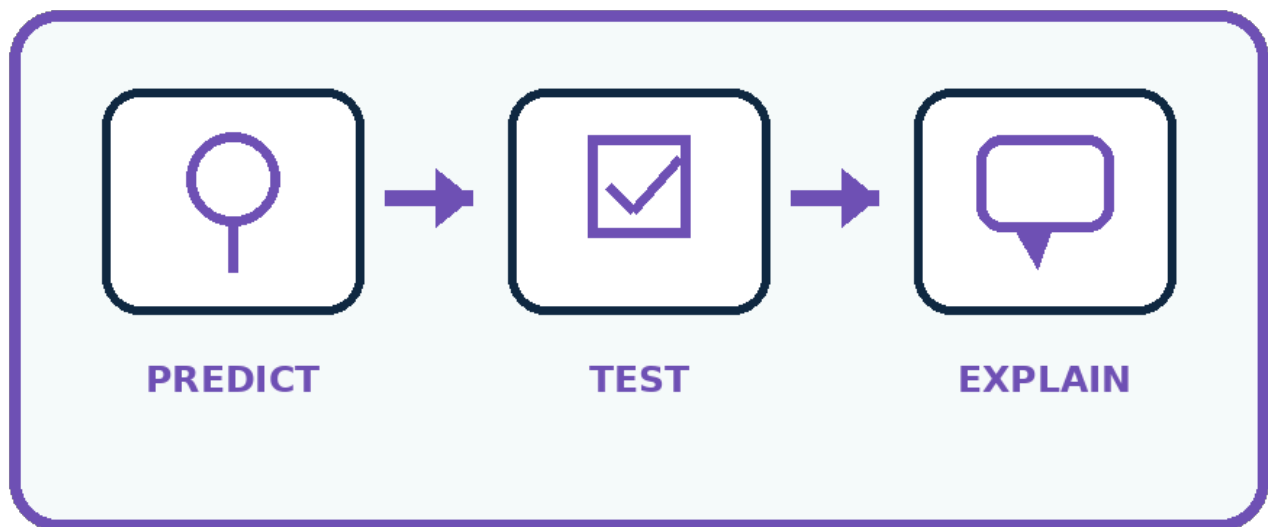
This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EXPLORER LAB

Pixel Picture Decoder

Predict first, test honestly, and explain what the evidence means.



State your prediction before the result is known.

Complete the task: Decode a tiny pixel picture and predict a blurred version.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

Case · Prediction · Observed result · What it means

Case: Normal

Prediction: _____

Observed result: _____

What it means: _____

Case: Edge

Prediction: _____

Observed result: _____

What it means: _____

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

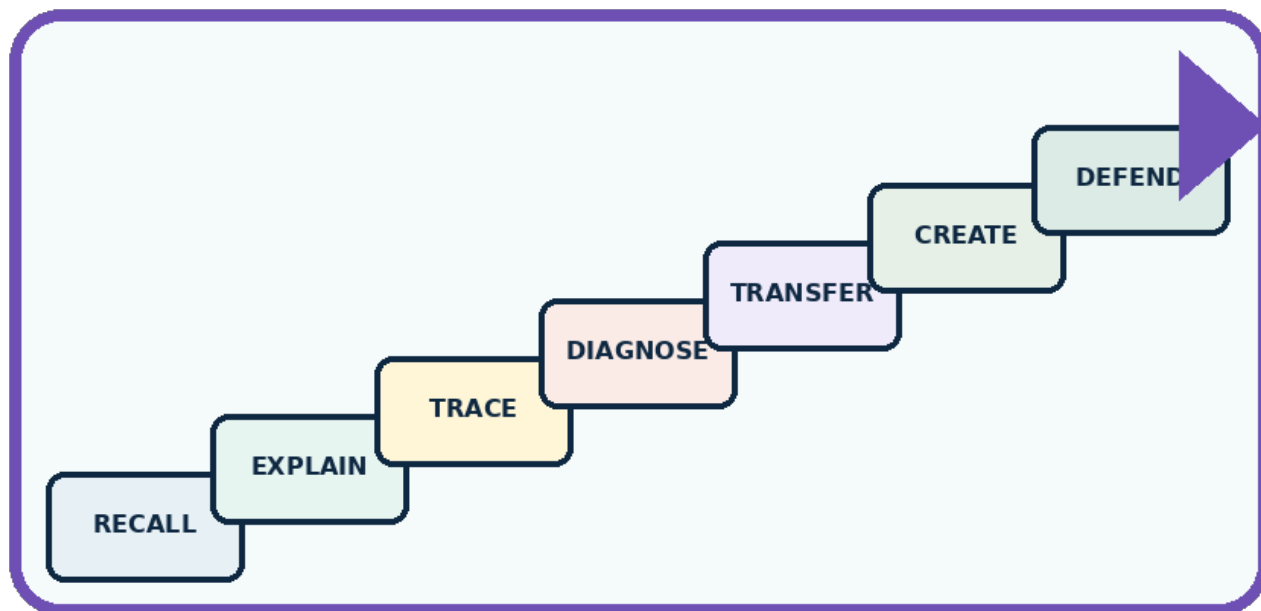
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define pixel.

EXPLAIN A digital image is a grid of measured color values.

TRACE Show the input, process, and output.

DIAGNOSE Repair: The computer stores a real cat inside the image file.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, evidence, mechanism, limit, and safe action.

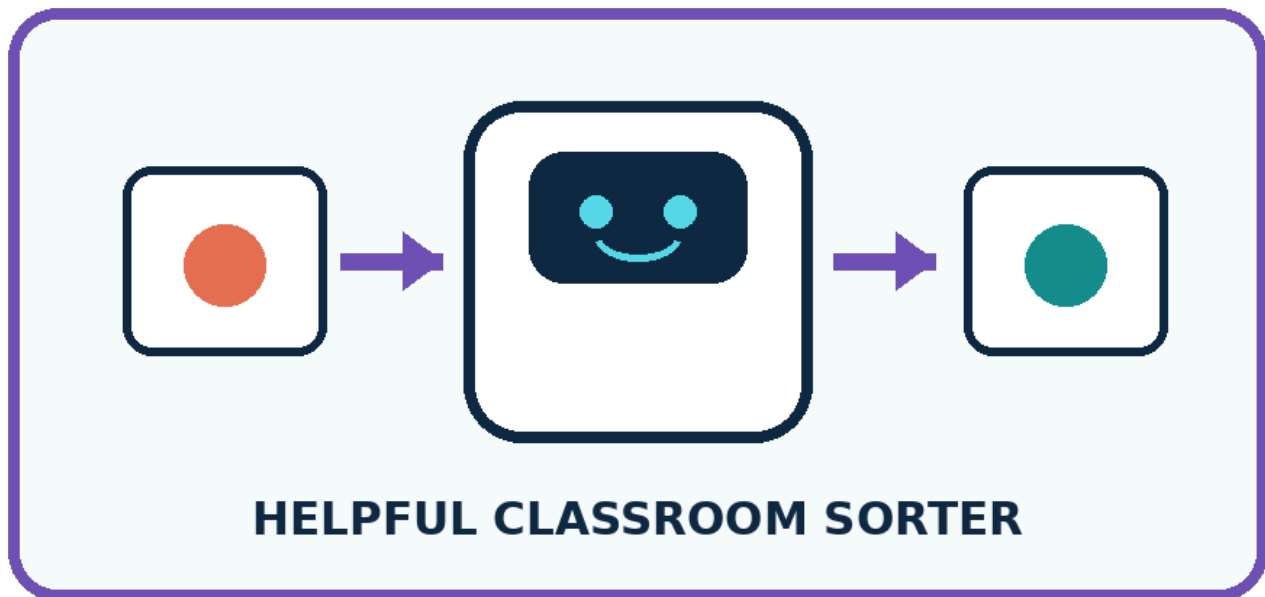
RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

11. Connect your project

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one evidence-based decision at a time.



THIS WEEK'S CHECKPOINT Choose which visible item features the Classroom Sorter may represent as image measurements and name one feature that lighting could distort.

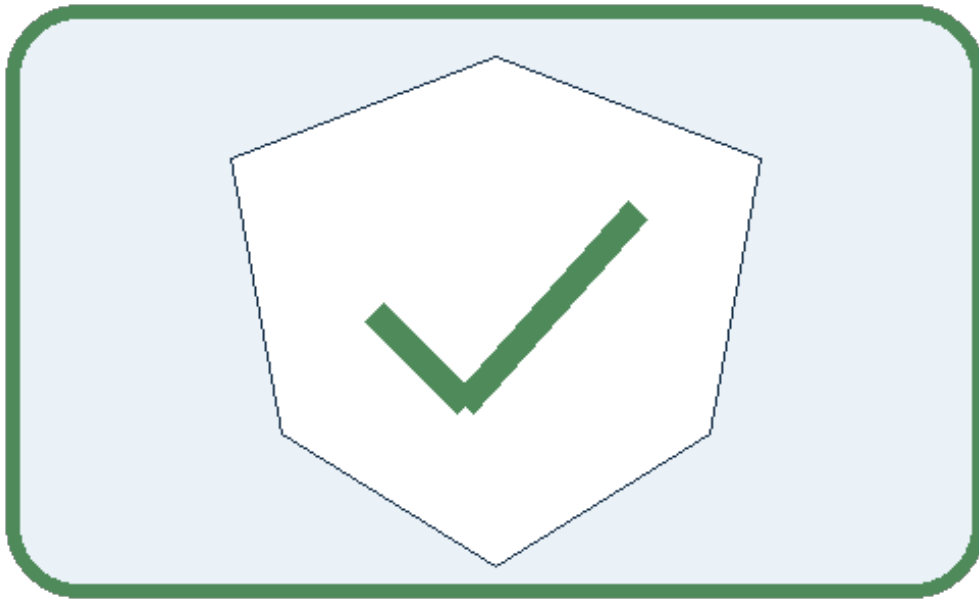
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the process?

What normal and edge cases will you test?

When should it say not sure or ask a person?



RESPONSIBLE DESIGN A pixel is a measurement at a location, not a tiny copy of the object.

My project decision:

Evidence for your project

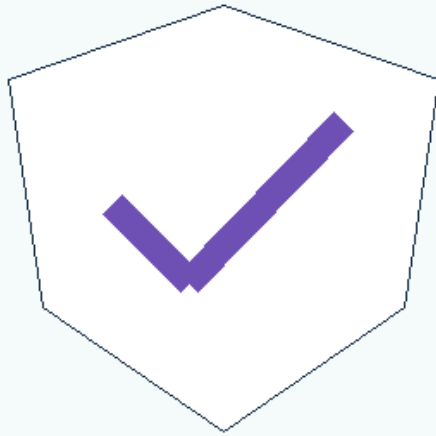
If the sorter uses images, retain dimensions and arrangement rather than assuming a dark-pixel count describes shape.

12. Show what you know

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



INDEPENDENT PROOF

REQUIRED PROOF Decode a 4 by 4 number grid, identify the pictured shape, then predict which details disappear when neighboring cells are averaged.

- ☐ Name or draw the fresh case.
- ☐ Trace input, process, and output.
- ☐ State the conclusion and strongest evidence.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent evidence:

SELF-CHECK Each grid position becomes one displayed square. A blur combines nearby values, so isolated corners and thin lines disappear before large regions. The computer transforms numbers; the human viewer recognizes the shape.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

NEXT BRIDGE Carry today's evidence habit into: Sound Becomes a Signal.

13. Study a complete case

Week 5 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This

case extends the illustrated lesson.

The situation

A 2 by 2 image has rows [0,1] and [1,0]. Zero means white and one means black.

The question

Draw it, then change the top-left value to 1.

Worked explanation

Initially top-right and bottom-left are black. The change produces three black squares. Pixel values encode an image; object recognition is a separate interpretation.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

A picture represented by four numbers

Evidence or stage · Value or result

Evidence or stage: Top left

Value or result: 0

Evidence or stage: Top right

Value or result: 1

Evidence or stage: Bottom left

Value or result: 1

Evidence or stage: Bottom right

Value or result: 0

A 2 by 2 image has rows [0,1] and [1,0]. Zero means white and one means black.

Draw it, then change the top-left value to 1.

Initially top-right and bottom-left are black. The change produces three black squares. Pixel values encode an image; object recognition is a separate interpretation.

14. Try a new case independently

Independent case: Design a new counterexample

Create two different images of width 4 and height 2, each with exactly three black pixels. Write both complete row encodings. Explain which fact their shared count retains and which difference it discards.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 5 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Use 0 for dark and 1 for light. Draw a two-by-two image with rows [1,0] and [0,1]. Then change only the bottom-left pixel to light. Which number changes?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Choose which visible item features the Classroom Sorter may represent as image measurements and name one feature that lighting could distort.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to

keep your practice evidence together.

Transfer practice from the original programme

Encode a 2×3 monochrome image with top row black–white–black and bottom row white–black–white, using 1 for black. A partner reads columns first. Explain why the image changes although the six values are identical.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Two fictional 2-by-3 pixel badges contain the same number of dark squares.

By the end, I can · How I will show it

By the end, I can: Encode and reconstruct a small image using an explicit grid convention

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Explain why pixel counts alone cannot recover an arrangement

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Distinguish numerical image data from a model's interpretation of that data

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Encode and reconstruct a small image using an explicit grid convention

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.